Pausing on the Answers: How Prequestions Guide Attention During Reading

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Author Note

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Abstract

Answering questions before a learning episode (prequestions) improves memory for the information the questions address, but why it does so is not well understood. One proposal is that prequestions guide attention toward the information they address. We tested this selective-attention account by recording eye movements while participants read a passage about efforts to eradicate polio and tuberculosis (TB). Before reading, participants answered prequestions about polio, prequestions about TB, or no prequestions; afterwards, all participants took a short-answer test on both topics. Prequestions produced a specific memory benefit—better memory than the no-prequestion condition for the prequestioned topic—and a small general cost for the non-prequestioned topic, for which the Bayes factor was inconclusive. Compared with the same sentences read without prequestions, sentences that answered a reader's prequestions received more occasional unusually long readings, were re-read more, and were more likely to be looked back to later, whereas typical reading time did not change. Non-prequestioned sentences were read largely as they were without prequestions. Prequestions thus guide attention selectively, but heightened attention is not the whole story: rather than slowing reading of the relevant material overall, they lead readers to pause on the answers and return to them, consistent with additional processing when readers recognize an answer to their prequestions.

Keywords: prequestions, pretesting, eye tracking, reading, attention, learning

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The idea that learning should be error-free and effortless, while intuitive, runs counter to decades of research at the interface of cognitive science and education. Making learning more effortful, and even more error-prone, can be highly beneficial for long-term retention (Bjork, 1994). This concept of *desirable difficulties* has inspired a large literature examining which learning techniques enhance memory and the mechanisms that underlie their benefits.

One of the most widely studied techniques is testing. Testing not only reveals gaps in knowledge to students and teachers, it also potentiates learning: retrieving studied information improves later memory more than restudying it does (S. K. Carpenter, 2012; Dunlosky et al., 2013; Kornell & Vaughn, 2016; Roediger & Karpicke, 2006; Rowland, 2014). This *testing effect*, or retrieval practice effect, is robust and has been found across many types of materials and contexts.

Attempting to retrieve information can also help *before* that information has been studied. In a typical prequestion study (e.g., Richland et al., 2009), learners are assigned to a prequestion group or a control group that receives no prequestions. The prequestion group first answers questions before a learning event and is encouraged to give its best guess even when the answers are unknown; the control group begins the learning event without answering any questions. After the learning event, both groups take a final test, which includes the questions that were prequestioned for the prequestion group and new, non-prequestioned questions. The typical finding is that the prequestion group remembers the prequestioned information better than the control group does—the *prequestion effect*. Because this benefit is specific to the information targeted by the prequestions, it is often called the *specific benefit* of prequestions (Pan & Carpenter, 2023). Prequestions sometimes also improve memory for information they did not address, a *general benefit*, but general benefits are typically smaller and appear to depend on the materials, such as short lecture videos (S. K. Carpenter & Toftness, 2017), on the kinds of questions used (Bostan & Ozcelik, 2026; St. Hilaire et al., 2019), and on where untested

information appears relative to the tested information (Sana & Carpenter, 2023). Two recent meta-analyses confirmed a robust specific benefit ($g = 0.54$, St. Hilaire et al., 2024; $g = 0.66$, King-Shepard et al., 2025) but found no reliable general benefit; in the larger, multilevel analysis, the general benefit was essentially zero ($g = 0.01$, King-Shepard et al., 2025). The specific benefit, in contrast, has been replicated with prose passages (Richland et al., 2009) and video-recorded lectures (S. K. Carpenter & Toftness, 2017; St. Hilaire & Carpenter, 2020), and has been examined in classroom settings, with mixed results (S. K. Carpenter et al., 2018; Geller et al., 2017; see Pan & Carpenter, 2023 for a review).

How exactly do prequestions benefit learning? Several mechanisms have been proposed —attention, elaboration, curiosity, and metacognition—and because they are not mutually exclusive, isolating any one of them is difficult (King-Shepard et al., 2025). They are also likely to interact: curiosity, for example, would likely increase attention to specific information, and becoming aware of a knowledge gap may make learners more curious and therefore more attentive. Attention may thus be a common route through which the other mechanisms operate, and the meta-analytic evidence aligns best with an attention-based explanation (King-Shepard et al., 2025). On this *selective-attention* account, prequestions signal what is likely to be important on an upcoming test, so learners invest more processing in the information the prequestions address—for example, by searching for the answers and dwelling on them when they find them —and that information is remembered better because it received more attention during learning (for a review of answer-search and attentional accounts, see Pan & Carpenter, 2023). A related attentional-window account adds that this heightened attention lasts only until the answers are found (Sana & Carpenter, 2023). On this account, the effect of prequestions is time-limited: once the answer to a prequestion has been found, attention is no longer heightened, so information that comes afterwards receives no general benefit. Sana and Carpenter (2023) tested this by manipulating where the prequestioned information appeared in the text. When it came at the

beginning, prequestions produced only a specific benefit, but when it came at the end, so that learners were still searching for the answers while reading the rest of the text, prequestions also produced a general benefit. Most of the evidence for these accounts, however, comes from what learners remember afterwards, not from how they process the material as they learn.

Eye tracking offers a direct way to test this account. It records where and for how long readers look while they read, without interrupting or otherwise altering reading, so that processing can be examined as it unfolds rather than inferred afterwards from a memory test (Rayner, 1998). Its interpretation rests on the eye–mind assumption: the information a reader is fixating is the information currently being processed, so that how long and how often readers look at a part of a text reflects how much processing it receives (Just & Carpenter, 1980; Rayner, 1998). Eye movements can also separate processing when information is first encountered from processing when readers later return to it. For these reasons, measures of real-time processing such as eye tracking have been identified as a key direction for prequestion research (King-Shepard et al., 2025; Pan & Carpenter, 2023). Few studies have used eye tracking to examine prequestions during reading. In an early study, readers given questions before reading re-read and processed the question-related material more carefully and read more quickly through unrelated material (Rothkopf & Billington, 1979). Lewis and Mensink (2012, Experiment 2) had students read a text about the humanitarian efforts of the Gates Foundation, which also described polio and tuberculosis (TB). One group read four prequestions about TB before reading (e.g., "How is tuberculosis spread through the air?"), and the other group read the passage without prequestions, while eye movements were recorded with a head-mounted eye tracker. Prequestions did not slow forward reading of the targeted sentences (e.g., "Tuberculosis is spread through the air when people who have the disease cough, sneeze, or spit"), but readers who received prequestions re-read them relatively more during first-pass reading, before moving on, and spent more time looking back to them after moving on to later parts of the passage. They

also looked back somewhat more to the rest of the text. On a free-recall test, the prequestion group recalled about twice as much of the targeted information as the control group, but no more of the other information. The specific benefit thus occurred without a general benefit, and it was accompanied by a shift of attention toward prequestioned information. Because only TB was prequestioned, however, the effect of prequestions could be measured only on the TB sentences, and the absence of a general effect only on the rest of the text, so neither could be separated from the content of those particular sentences; the comparisons were also between two groups of about 21 readers, based on average reading times.

More recently, Bostan and Ozcelik (2026) gave 12 adults *integrative* prequestions—questions that require combining information from different parts of a text—before they read a passage about car brakes, and compared them with 12 adults who read without prequestions. The prequestion group made more fixations and spent more time on prequestioned sentences, and made more transitions between them, than the control group. Unlike most prequestion studies, they also found a general benefit: the prequestion group scored higher on non-prequestioned questions, even though they spent less time on non-prequestioned sentences. These findings are consistent with an attentional account, but the design had the same limitation as that of Lewis and Mensink (2012): one fixed set of sentences was prequestioned, so the effects could not be separated from the content of those sentences. In addition, with two prequestions and 12 readers per group, compared on participants' total fixation times and counts, the estimates are imprecise, and differences between the groups in how they read the same sentences cannot be separated from differences between the readers themselves.

Whether attention is the main mechanism is not settled, however. Pretesting has been found to reduce mind-wandering during online lectures (Pan et al., 2020), but in a larger study, content-matched pretesting neither reduced mind-wandering nor improved learning, and interpolated testing, although it slightly reduced mind-wandering, did not improve learning either

(Welhaf et al., 2022). In a large multi-site classroom study, prequestions before online lecture videos improved learning even though students did not watch more of the videos afterwards (Motz et al., 2026), suggesting that prequestions can improve learning without increasing the time spent on the material. Motz et al. (2026) proposed instead a *mental model hypothesis*: prequestions activate relevant knowledge before learning, which then facilitates the processing and encoding of related information. On this view, attention effects follow from the activated knowledge: when learners encounter content related to the prequestions, it resonates with the activated knowledge, and once they recognize it as related, they may direct additional attention to it.

The Current Study

The current study tested the selective-attention account with a design that addresses the limitations of the two previous eye-tracking studies of prequestions during reading (Bostan & Ozcelik, 2026; Lewis & Mensink, 2012). Participants read the Gates Foundation text used by Lewis and Mensink (2012), which discusses efforts to eradicate two diseases, polio and tuberculosis (TB). First, unlike in previous studies, the prequestioned content was counterbalanced: two prequestion groups were prequestioned on different topics, so that the effect of prequestions could be separated from the content of the sentences. Second, the sample was substantially larger—138 participants, about 46 per condition—and the models included variation across sentences and test questions as well as across readers, so that the effects generalize beyond these particular items. Third, beyond average reading times, we modelled the distribution of reading times, which distinguishes slower reading overall from occasional unusually long readings. Finally, unlike the integrative prequestions of Bostan and Ozcelik (2026), each prequestion was answered by a single sentence of the text, as in Lewis and Mensink (2012), the kind of prequestion that typically produces a specific but not a general benefit (St. Hilaire et al., 2019). Before reading, participants answered five prequestions about polio, five

prequestions about TB, or no prequestions (no-prequestion condition). Each of ten target sentences in the text contained the answer to one question. Because the Polio and TB conditions swapped which topic was prequestioned, every target sentence served as prequestioned information for one group of readers and as non-prequestioned information for another. Pooling the two prequestion conditions, reading of prequestioned sentences could therefore be compared with reading of the very same sentences without prequestions (the no-prequestion condition), with each topic contributing equally.

We examined two kinds of eye-movement measures. *Global* measures—total gaze duration, re-reading time, and second-pass regressions—reflect all processing a sentence receives, including later returns to it. The *local* measure—first-pass duration—reflects processing when a sentence is first encountered. Distinguishing the two indicates *when* prequestions change processing: as readers first encounter the answer, or when they return to it later.

Fixation and reading times are not normally distributed: they cannot fall below zero and are strongly right-skewed, with a long tail of unusually long readings (Balota & Yap, 2011; Staub et al., 2010). Analyses that assume normality therefore misdescribe them, and transforming the data to reduce the skew changes the hypotheses being tested (Lo & Andrews, 2015). We therefore analysed total gaze duration with the ex-Gaussian distribution, which describes durations as the sum of a Gaussian component, with mean μ, and an exponential component, with mean τ, and so models the skew directly. Beyond fitting the shape of the data, the ex-Gaussian has a second advantage: it separates two ways in which reading times can change, a shift of the whole distribution (μ) and a lengthening of its tail (τ), which an analysis of means cannot tell apart. Although the ex-Gaussian is a descriptive model of response times, its parameters have often been mapped onto cognitive processes, though these mappings remain debated (Matzke & Wagenmakers, 2009). The Gaussian component (μ) is typically linked to relatively automatic, early-stage processing, whereas the exponential component (τ) is linked to

later, more controlled processes such as attentional control, working memory updating, error monitoring, and strategic processing. In eye-tracking research on reading, word frequency affects τ as well as μ of fixation durations (Staub et al., 2010), whereas a modest degradation of the text, such as presenting it faintly, shifts only μ (White & Staub, 2012); in scene viewing, the type of image shifts μ, whereas familiarity with the image lengthens only a subset of relatively long fixations, increasing τ (Guy et al., 2020). Some manipulations, then, lengthen a subset of unusually long fixations rather than shifting the whole distribution. Previous eye-tracking studies of prequestions analysed average reading or fixation times (Bostan & Ozcelik, 2026; Lewis & Mensink, 2012; Yang et al., 2021). Averages can mask effects of this kind: if prequestions add processing to only some encounters with a sentence, the mean changes little, and a longer tail cannot be distinguished from slower reading overall (Guy et al., 2020). Here, sentences that readers recognize as answering a prequestion may trigger reanalysis or additional processing. If so, prequestion effects should appear primarily in τ rather than μ. Because μ and τ are both parameters of the ex-Gaussian model, we examined both: an effect on μ would indicate that prequestions shift typical reading of the prequestioned sentences as a whole, whereas an effect on τ would indicate that they lengthen only a subset of readings, the unusually long ones, consistent with additional processing on some encounters.

Hypotheses

For final test performance, we compared memory for prequestioned and for non-prequestioned material with memory for the same material in the no-prequestion condition, overall and on each topic. We predicted a specific benefit—better performance than the no-prequestion condition on the prequestioned topic—but no general benefit on the non-prequestioned topic.

For eye movements, we predicted a specific effect: prequestioned information would receive longer gaze durations and longer re-reading than non-prequestioned information and than

the same information read without prequestions (the no-prequestion condition). Broken down by topic, readers given Polio prequestions should spend longer on Polio than on TB sentences, readers given TB prequestions should show the reverse pattern, and no-prequestion readers should show no difference between the two topics.

Method

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study. The study was preregistered (https://osf.io/8env4). All data, materials, and analysis scripts and code are available at https://github.com/jgeller112/Prequestions_EyeTracking; every number, table, and figure in this article is computed from the raw data by the code in that repository.

Participants

The target sample size was based on the specific benefit of prequestions. A recent meta-analysis reported a medium-sized specific benefit ($g = 0.54$; St. Hilaire et al. (2024)), corresponding to an odds ratio of 2.66. A simulation-based power analysis with mixedpower (Kumle et al., 2021) for the preregistered binomial mixed model, based on pilot data, indicated that 45 participants per group would provide 90% power to detect an effect of this size. We therefore planned to test 45 participants in each of the three conditions (135 in total).

Participants were recruited from the Boston College SONA subject pool and received course credit. Eye-tracking data were recorded from 161 participants. One participant had eye-tracking data for fewer than all four pages of the text and was excluded. Eighteen participants did not meet the preregistered calibration criterion (accuracy and precision of 0.5° or better) or could not be calibrated and were excluded. Four participants had no final test data and were excluded. The final sample comprised 138 participants (no prequestions: 47; Polio prequestions: 45; TB prequestions: 46). Calibration validation data were available for all included participants. Mean accuracy was 0.37° ($SD = 0.10$, range 0.13–0.50), and mean precision (SD of gaze samples) was

0.22° (*SD* = 0.12, range 0.06–0.49). All included participants with calibration data met the criterion. The preregistered exclusion of participants with high prior knowledge was not applied.

Materials

Participants read a 690-word, multi-topic text (Lewis & Mensink, 2012) that discussed the Gates Foundation's humanitarian efforts in eradicating Polio and TB in the developing world. The text consisted of 38 sentences presented over four pages, each 16–17 lines long (Figure 1). Ten sentences served as target sentences: five about Polio (*M* = 19.4 words, *SD* = 11.4, range 4–35) and five about TB (*M* = 16.2 words, *SD* = 3.7, range 11–21). The Polio target sentences had a higher reading level than the TB target sentences, as measured by the Flesch–Kincaid grade level (Kincaid et al., 1975), computed with quanteda (Benoit et al., 2018): Polio *M* = 14.1, *SD* = 6.2; TB *M* = 8.8, *SD* = 2.0. Their syntactic complexity, estimated from a dependency parse with UDPipe (Straka & Straková, 2017; Wijffels & Straka, 2026) and its model of the English Web Treebank (Silveira et al., 2014), was similar: mean dependency distance (Liu, 2008), the average number of words between each word and its syntactic head (Polio *M* = 2.72, *SD* = 1.03; TB *M* = 2.62, *SD* = 0.40), and number of clauses per sentence (Polio *M* = 1.6, *SD* = 0.9; TB *M* = 1.8, *SD* = 0.8). Each target sentence contained the answer to one of ten short-answer questions (five Polio, five TB; Appendix A). Questions were percentage-based, yes/no, or multi-part, and were worth 12 points per topic (24 points in total).

There were three conditions. In the Polio prequestion condition, participants answered the five Polio questions before reading; in the TB prequestion condition, they answered the five TB questions; in the no-prequestion condition, they read the text without prequestions. For each prequestion condition, its five target sentences were *prequestioned* and the other topic's five were *non-prequestioned*, so every target sentence served in both roles across conditions.

Apparatus

Eye movements were recorded with a Tobii Pro Spectrum eye tracker sampling at 600 Hz. The text was displayed on an EIZO FlexScan EV2451 monitor (23.8-inch, 1920 × 1080

pixels) viewed from approximately 65 cm, without a chin rest. The text was displayed in 36-point Courier New, a monospaced font, on a grey background. At this distance, one character subtended approximately 0.5° of visual angle and one line spacing approximately 1.1°.

The whole experiment was run in Tobii Pro Lab (version 25.19). The prequestions, distractor task, and final test were PsychoPy tasks hosted on Pavlovia and presented as web elements within the Tobii Pro Lab timeline.

Procedure

At the start of the session, each participant was randomly assigned to one of the three conditions (Polio prequestions, TB prequestions, or no prequestions). The session had four phases—prequestions, reading, a distractor task, and the final test—and participants in the no-prequestion condition skipped the first.

Prequestions

Participants in the two prequestion conditions answered the five short-answer questions on their topic (Polio or TB), in a random order (https://run.pavlovia.org/Jgeller112/preq). They were told that they might not know the correct answers but should give their best guess. Each question was shown on the screen with a box below it, into which participants typed their response. They had unlimited time and pressed a triangle at the bottom of the screen when they were finished. No feedback was given.

Reading

Before reading, the eye tracker was calibrated with Tobii Pro Lab's 9-point calibration procedure (white targets on a grey background), and calibration accuracy and precision were validated and recorded. Participants then read the text, presented over four pages, while their eye movements were recorded.

Distractor Task

After reading, all participants completed the president recognition task (https://run.pavlovia.org/Jgeller112/preq_president), a distractor task lasting approximately 3 minutes.

Final Test

Participants then answered all ten final-test questions—five on Polio and five on TB—in a random order (https://run.pavlovia.org/Jgeller112/preQ_test), in the same format as the prequestions, with unlimited time and no feedback. For participants in the prequestion conditions, five of the questions were the prequestions they had answered before reading, and five were new. Finally, participants were thanked and debriefed.

Scoring

Prequestion and final test responses were scored by Claude Opus 5.5 (Anthropic; model ID `claude-opus-5-5`). Rather than judging responses one at a time, Claude wrote a deterministic, rule-based scoring script (`scoring_rubric.R`, with `grade_prequestions.R` and `grade_test.R`) that awarded one point for each element of the correct answer present in a response, and applied it identically to every response. The scoring rules for each question are given in Appendix A. To check the automated scoring, a research assistant who was blind to condition, test phase, and the automated scores independently scored all 1,665 responses that were not blank. The two scorings agreed exactly on 97.6% of responses, and agreement was excellent, ICC(2,1) = .99, 95% CI [.98, .99], using a two-way random-effects, absolute-agreement, single-rater ICC (Koo & Li, 2016; Shrout & Fleiss, 1979). For participants' final test totals, ICC(2,1) = .99. Agreement for each question is given in Table A2. All analyses used the automated scores.

Eye-Movement Data Processing

Sentence-level measures were computed from the fixation-level data rather than taken from Tobii Pro Lab's sentence metrics, for two reasons. First, when a page appeared, readers' eyes typically landed partway down the page before moving to the first line, (on 79% of pages, the first fixation on the text fell below the first sentence), so Tobii Pro Lab treated the sentences above the landing point as already passed and recorded no first-pass reading for them. Second,

recorded gaze positions were often offset vertically by up to about half a line, so fixations were sometimes assigned to the sentence on a neighbouring line.

Fixations were identified with Tobii Pro Lab's Tobii I-VT (Fixation) filter, a velocity-threshold identification algorithm (Olsen, 2012; Salvucci & Goldberg, 2000), with the following settings. Gaps in the gaze data shorter than 75 ms (e.g., brief losses of tracking) were filled by linear interpolation, the gaze positions of the two eyes were averaged, and noise was reduced with a moving median over a window of three samples. A gaze sample was classified as part of a fixation when the angular velocity of the eye, computed over a 20-ms window, was below 30°/s. Adjacent fixations separated by less than 75 ms and less than 0.5° of visual angle were merged, and fixations shorter than 60 ms were discarded. Fixations cut off when a page appeared or disappeared can nonetheless be shorter than 60 ms in the export. We excluded fixations shorter than 80 ms (4.0% of fixations) and longer than 1,000 ms (0.06%) before any further processing. Both bounds follow common practice in reading research: 80 ms is the most common lower bound, and 800, 1,000, and 1,200 ms are the most common upper bounds (Schotter & Dillon, 2025; e.g., Serrano-Carot et al., 2025). Very short fixations are unlikely to reflect processing of the text and include fixations cut short by a page change. Most fixations in reading last around 200 ms (Rayner, 1998; Schotter & Dillon, 2025); in the present data the median fixation lasted 178 ms, so a single fixation longer than 1,000 ms, 5.6 times the median, is unlikely to reflect ongoing reading and more likely reflects a lapse of attention or a brief loss of the eye-tracking signal that the fixation filter bridged.

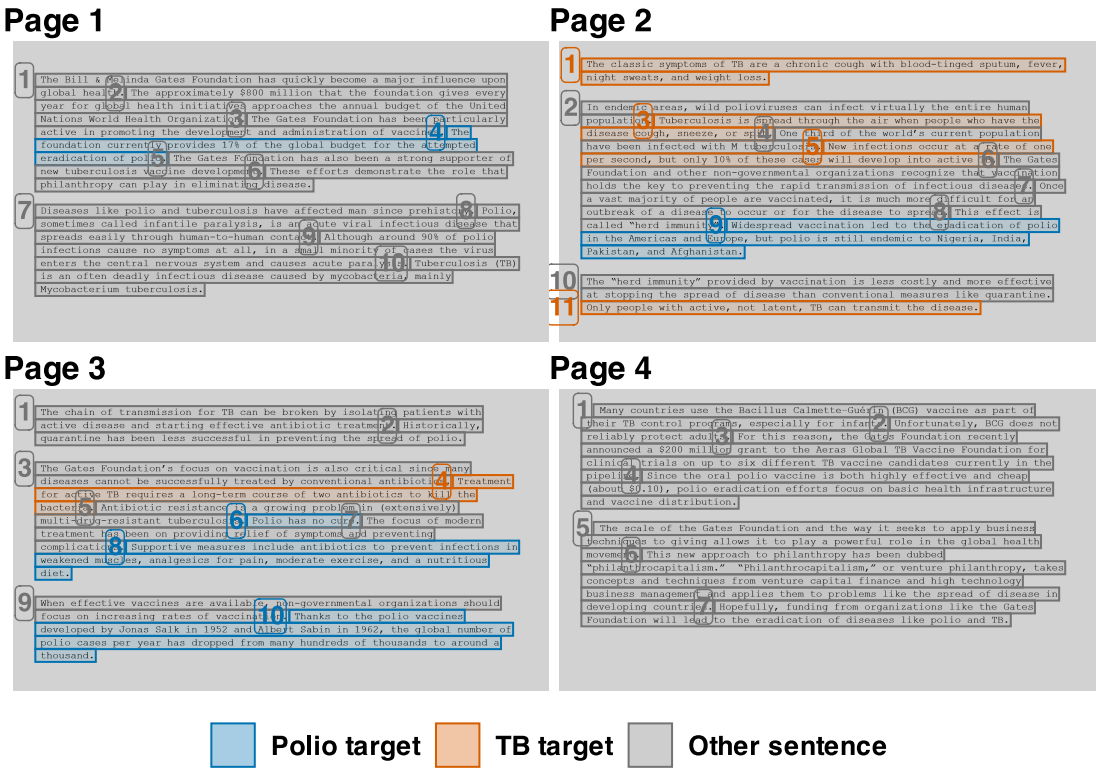
The remaining processing (`derive_et_measures.R`) proceeded in four steps. (a) The exact layout of each page was reconstructed from the sentence texts and full-resolution screenshots of the pages (monospaced font, 79 characters per line), giving the screen position of every character of every sentence (Figure 1). (b) Each fixation was assigned to a text line after correcting for vertical drift, separately for each reader and page. The correction combines two

approaches used in automated drift-correction algorithms (Carr et al., 2022): assigning sequences of fixations to a line together, and modelling drift as a function of position on the page. First, a single vertical offset was estimated: candidate shifts of the recorded positions were tried in steps of 0.05 line spacings, and the shift that minimized the summed squared distance between each fixation and its nearest text line was kept (distances were capped at one line spacing, so that stray fixations in the margins had little influence). Second, consecutive fixations moving along a line were grouped into chains; a chain ended at a horizontal jump larger than 10% of the screen width (e.g., a return sweep to the next line) or a vertical jump larger than 0.6 line spacings. Each chain was assigned as a whole to the text line nearest its median position, so that a single fixation drifting toward a neighbouring line stayed on the line being read. Third, because drift often grew down the page or across it, a linear model of the recorded vertical position as a function of the assigned line's position and the fixation's horizontal position (recorded $y = a + b \times \text{line } y + c \times x$) was fitted to fixations whose chain lay within half a line spacing of its line; gaze positions were corrected by inverting this model and the chains were reassigned, and this was repeated five times. The linear model was used only if it was plausible (a vertical stretch within 20% and a tilt of less than one line across the screen); otherwise the single offset was kept. Throughout, the correction was limited everywhere on the page to the calibration accuracy criterion of 0.5° (0.57 cm, about 0.45 of a line spacing), so it could not move a reader's gaze by a whole line. Chains more than half a line spacing from every text line were treated as off the text. (c) Each fixation was then assigned to the sentence occupying its line and horizontal position: its character column was computed from the fixed character width, a fixation on a space took the sentence of the nearest letter, and fixations more than three characters beyond either end of a line were treated as off the text. (d) Reading on each page was taken to start at the first fixation on the page's first sentence; fixations before that (the landing) contributed to total time but not to first-pass measures. After processing, 96.8% of on-screen fixations were assigned to a sentence, and the

assigned sentence agreed with Tobii Pro Lab's assignment for 89.8% of fixations. First-pass reading was identified for 96.9% of fixated target sentences. Figure 2 shows one reader's fixations on one page before and after processing. For this reader, the correction moved fixations by a median of 0.44 line spacings and changed the sentence assigned to 43.4% of fixations; the medians over all readers and pages were 0.24 line spacings and 8.2%.

Figure 1

The Four Text Pages and the Sentence Regions Used as Areas of Interest



Note. Each sentence is one area of interest, made of one rectangle per text line it occupies. Polio target sentences are shown in blue, TB target sentences in vermillion (orange-red), and other sentences in grey; numbers give each sentence's position on its page. Page 4 contains no target sentences.

Figure 2

One Reader's Fixations on One Page Before and After Processing



Note. One participant in the TB prequestion condition, page 2, chosen to illustrate vertical drift. (A) Recorded gaze positions, coloured by the sentence assigned by Tobii Pro Lab. (B) The same fixations after drift correction (single vertical offset, assignment of chains of fixations to lines,

and a linear drift model, limited to 0.5°; see Eye-Movement Data Processing), coloured by the sentence assigned in processing. Each point is one fixation (size proportional to duration); grey points were not assigned to any sentence. Lines join consecutive fixations. Open circles are fixations made before reading started on the page. Dashed rectangles outline the target sentences.

Eye-Movement Measures

All measures were computed for each participant and target sentence. Durations were divided by the number of words in the sentence (ms per word), because target sentences differed in length. Only sentences that were fixated were analysed: a target sentence that received no fixation has no reading time, so its measures were treated as missing (17 of 1380 target sentences, 1.2%). Following the eye-movement literature, we distinguish *global* measures, which reflect all processing of a sentence including later returns to it, from *local* measures, which reflect processing when the sentence is first encountered (Table 1).

Table 1

Eye-Movement Measures

Measure	Type	Definition	Analysed as
Total gaze duration	Global	Summed duration of all fixations on the sentence.	Ex-Gaussian: μ (typical reading time) and τ (tail of unusually long readings)
Re-reading time	Global	Summed duration of fixations on the sentence after its first pass, that is, on returns to it (total gaze duration minus first-pass duration, excluding fixations before reading started); 0 for sentences that were not re-read. Operationalizes the preregistered re-reading measure.	Hurdle Gamma: whether the sentence was re-read, and re-reading time when it was; reported as the expected re-reading time per sentence
Second-pass regression (look-back)	Global	Whether the reader returned to the sentence from later text after first-pass reading had moved past it (0/1).	Logistic
First-pass duration	Local	Summed duration of fixations from first entering the sentence until first leaving it, provided no later sentence had been read.	Gamma (log link)

Note. All measures were computed for each participant and target sentence. Durations were divided by the number of words in the sentence and are reported in milliseconds per word. A *pass* is a sequence of consecutive fixations on the same sentence. First-pass duration is defined only for sentences read during first-pass reading (not skipped). Global measures include later returns to the sentence; the local measure reflects processing when the sentence is first encountered.

Statistical Analysis

All analyses were conducted in R version 4.6.1 (R Core Team, 2026).[1] The effects of prequestions on test performance and eye movements were estimated with Bayesian mixed models in brms (Bürkner, 2017), which fits models in Stan (B. Carpenter et al., 2017), run through cmdstanr (Gabry et al., 2025) with CmdStan 2.37.0. Each target sentence, and each final-test question, was coded by its status for the reader: *prequestioned* (Polio sentences and questions for readers in the Polio condition, TB sentences and questions for readers in the TB condition), *non-prequestioned* (the other topic for those readers), or *no-prequestion* (both topics for readers in the no-prequestion condition). The main models pooled the two prequestion conditions, with Status and Topic (Polio, TB) as fixed effects. The *specific effect* of prequestions was the difference prequestioned – no-prequestion: the same sentences or questions for readers whose prequestions they answered and for readers who saw no prequestions. The *general effect* was non-prequestioned – no-prequestion. Because each topic contributes equally to the prequestioned, non-prequestioned, and no-prequestion cells, differences between the Polio and TB materials cancel in these comparisons. Both the specific and general effects and their breakdown by topic were preregistered. To break the effects down by topic, we fitted the preregistered Group (Polio prequestions, TB prequestions, or no prequestions) × Topic (Polio, TB) models with the same families and priors. Following the preregistered hypotheses, we estimated the Polio – TB difference within each group—predicted to be positive in the Polio group, negative in the TB group, and absent in the no-prequestion group—and the Group × Topic interaction contrast, the Polio group's Polio – TB difference minus the TB group's, predicted to

[1] Data were processed and analysed in R version 4.6.1 (R Core Team, 2026) with the packages tidyverse 2.0.0 (Wickham et al., 2019), janitor 2.2.1 (Firke, 2024), png 0.1.9 (Urbanek, 2026), modelbased 0.16.0 (Makowski et al., 2025), marginaleffects 0.32.0 (Arel-Bundock et al., 2024), insight 1.5.2 (Lüdecke et al., 2019), brms 2.23.0 (Bürkner, 2017), cmdstanr 0.9.0 (Gabry et al., 2025), cogmod 0.3.3 (Makowski, 2026), ordbetareg 0.8 (Kubinec, 2025), bayestestR 0.18.1 (Makowski, Ben-Shachar, & Lüdecke, 2019), priorsense 1.4.0 (Kallioinen et al., 2023), ggdist 3.3.3 (Kay, 2024), quanteda 4.5.0 (Benoit et al., 2018), quanteda.textstats 0.97.2 (Benoit et al., 2024), udpipe 0.8.16 (Wijffels & Straka, 2026), patchwork 1.3.2 (Pedersen, 2025), scales 1.4.0 (Wickham et al., 2025), knitr 1.51 (Xie, 2015), and xfun 0.60 (Xie, 2026). The manuscript was produced with Quarto 1.10.18 and the apaquarto extension.

be positive. Because the Polio and TB sentences (and questions) are different items, each within-group Polio – TB difference combines any effect of prequestions with differences between the items, and with five items per topic, the population-level topic difference is estimated imprecisely. Each within-group difference was therefore interpreted together with the interaction contrast, in which the item differences cancel because both groups read the same items.

Reading times are strongly right-skewed, so no duration was analysed with a Gaussian model. As preregistered for heavily skewed durations, total gaze duration was analysed with an ex-Gaussian distributional model (cogmod; Makowski (2026)), which describes a duration as the sum of a Gaussian component, whose mean μ reflects typical reading time, and an exponential component, whose mean τ reflects the tail of unusually long readings. The fixed and random effects applied to both μ and τ, and σ had by-participant random intercepts. The ex-Gaussian could not be estimated for first-pass duration and re-reading duration: both include a large cluster of very brief readings (single short fixations on a sentence), so their distributions have no Gaussian component and its spread σ collapsed to zero. First-pass duration was therefore analysed with a Gamma model with a log link, which models positive, right-skewed durations without transforming them (Lo & Andrews, 2015). Re-reading duration was 0 for the 38.1% of fixated target sentences that were never re-read, which a Gamma distribution cannot produce, so it was analysed with a hurdle Gamma model. The model has two parts with the same predictors and random effects: a logistic part for whether a sentence was re-read, and a Gamma part (log link) for how long it was re-read when it was. The reported re-reading duration combines the two: the expected re-reading time per sentence, including the sentences that were not re-read. Second-pass regressions (0/1) were analysed with logistic models. Because the questions were worth different numbers of points (1–4), final test performance was analysed as the points earned on each question out of its maximum, with a beta-binomial model: a binomial model of the points earned with an additional parameter for overdispersion, because a participant tends to

know all or none of a question's elements. Effects are reported on the scale of the expected proportion of a question's points, which puts all questions on the same 0–1 scale. Because each question has at most four points, scores can take only a few values (0, .25, .33, .50, .67, .75, and 1). In posterior predictive checks, the beta-binomial model reproduced how often each score occurred (predicted and observed shares within 1.1 percentage points), better than a binomial model without overdispersion and an ordered beta model of the proportion of points (Geller et al., in press; Kubinec, 2023), which placed 29% of its predicted scores between the possible values. Prequestion performance was analysed in the same way, comparing the two prequestion conditions.

All models included the maximal random-effects structure warranted by the design (Barr et al., 2013). Topic varied within participants, so the models included by-participant random intercepts and Topic slopes (the preregistered structure). Every target sentence and every final-test question was prequestioned for one group, non-prequestioned for another, and read or answered by the no-prequestion group, so Status (and Group) varied within items, and the models also included by-sentence (eye movements) or by-question (final test) random intercepts and Status (or Group) slopes. Within a participant, Status is determined by Topic, so no by-participant Status slope was possible. In the prequestion model, condition varied between participants and between questions, so it included random intercepts for participants and questions only. Status, Condition, and Topic were coded with equal-prior contrasts [`contr.equalprior_pairs`; bayestestR, Makowski, Ben-Shachar, and Lüdecke (2019)]. With this coding, a normal(0, s) prior on the coefficients implies the same normal(0, s) prior on every pairwise difference between conditions, so that no condition serves as a reference with a different prior, and every comparison is tested against the same prior. s was set on each model's link scale: 1.5 on the log-odds scale (second-pass regressions, final test, and prequestions; an odds ratio of 4.5 for one *SD*), 0.5 on the log scale of the Gamma models and on the scale of τ (a factor of

about 1.65 for one *SD*), and 100 ms per word for μ. Intercepts, random-effect standard deviations, and correlations had weakly informative priors (Appendix B). Each model was sampled with four chains of 10000 post-warm-up iterations (40000 posterior draws, enough for stable Bayes factors). All models converged: there were no divergent transitions, all R-hat values were below 1.001, and all bulk effective sample sizes exceeded 5400.

Means and comparisons were computed on the response scale by averaging each posterior draw's predictions over the observed sample (modelbased with the marginaleffects backend and `estimate = "average"`), so that they incorporate the variation due to the random effects. Durations are reported in ms per word, and probabilities and proportions in percent. Effects were summarized by their posterior mean, 95% credible interval (CrI), and probability of direction (pd), the proportion of the posterior with the same sign as its median. The pd is closely related to the frequentist p value: for a one-tailed test, $p \approx 1 - \text{pd}$, and for a two-tailed test, $p \approx 2(1 - \text{pd})$, so pd > 95% corresponds to a one-tailed $p < .05$ (Makowski, Ben-Shachar, Chen, et al., 2019). The predictions for the specific effects and for prequestioned versus non-prequestioned material were directional (longer reading, more regressions, better recall), so we treated such an effect as present when pd > 95% in the predicted direction. The general effects had no directional prediction, so for them we required pd > 97.5%, corresponding to a two-tailed $p < .05$. Bayes factors were not used as a decision criterion; we report them to describe the strength of the evidence for or against each effect. Because pd only indicates the direction of an effect and cannot provide evidence that an effect is absent, we also computed Bayes factors. Each Bayes factor compares a model in which the comparison equals 0 with one in which it is free, using the Savage–Dickey density ratio (Wagenmakers et al., 2010): the posterior density of the comparison at 0 divided by its prior density at 0. The prior densities came from the same models sampled from their priors only, and both densities were computed for the comparisons on the link scale, where the equal priors apply. For τ and for the expected re-reading time, which

have no single link scale, they were computed on the response scale; for the expected re-reading time, at the population level (without random effects), because averaging over random effects drawn from their priors produced prior draws too extreme for a density estimate. For the Group × Topic comparisons of look-backs and final test scores, the Bayes factors were also computed on the response scale (probabilities and proportions of points), the scale of the estimates. These comparisons involve different Polio and TB items, some of which had outcomes near 0 or 1, and for such outcomes the same comparison is much less precise in log-odds than in probability, so link-scale Bayes factors could contradict the estimates. $BF_{10} > 3$ indicates moderate, $BF_{10} > 10$ strong, $BF_{10} > 30$ very strong, and $BF_{10} > 100$ extreme evidence for a difference, and $BF_{01} = 1/BF_{10}$ the corresponding evidence for no difference; values between 1/3 and 3 are anecdotal (Lee & Wagenmakers, 2014). Because a Bayes factor depends on the prior, all comparisons between conditions received the same prior, and we report how sensitive the posteriors were to the priors [power-scaling sensitivity; Kallioinen et al. (2023)].

Deviations From the Preregistration

(a) The preregistered re-reading measure was operationalized as second-pass time, because Tobii Pro Lab's re-reading duration (go-past time minus first-pass time) mostly reflects time spent on earlier text rather than on the target sentence. (b) Measures were computed from the fixation data rather than taken from Tobii Pro Lab (see Eye-Movement Data Processing). (c) Durations were standardized by sentence length. (d) The preregistration specified frequentist mixed models. All confirmatory comparisons were instead estimated with Bayesian models and summarized by the probability of direction and Bayes factors, which can also quantify evidence for the absence of an effect. (e) The models added by-sentence (eye movements) or by-question (final test) random intercepts and Status (or Group) slopes to the preregistered by-participant intercepts and Topic slopes, giving the maximal structure warranted by the design. (f) First-pass duration and re-reading duration were analysed with Gamma and hurdle

Gamma models rather than the preregistered linear mixed model, because the preregistered alternative for skewed durations, the ex-Gaussian, could not be estimated for them. (g) Final test and prequestion performance were analysed as the points earned on each question out of its maximum with beta-binomial models, rather than as points per topic. (h) The preregistered specific and general effects (prequestioned and non-prequestioned material versus the no-prequestion condition) were estimated with models that pooled the two prequestion conditions (Status and Topic as fixed effects), in addition to the preregistered Group × Topic models; in the Group × Topic models, the preregistered within-condition Polio – TB contrasts are reported together with the interaction contrast (see Statistical Analysis). (i) The preregistration listed first-pass regressions as a measure; this was an error for second-pass regressions (whether readers returned to a sentence from later text), the measure we intended and report. First-pass regression is therefore not reported; its analysis is included in the analysis code. (j) We planned to test 45 participants per condition (135 in total). After exclusions, the final sample comprised 138 participants, and the conditions were not exactly equal in size (no prequestions: 47; Polio prequestions: 45; TB prequestions: 46).

Results

Prequestion Performance

As intended, participants knew little of the material before reading. Participants in the Polio prequestion condition earned 4.1% of the available points on the prequestions (*SD* = 5.2%) and those in the TB prequestion condition 10.5% (*SD* = 10.9%). Participants attempted 43.1% (Polio) and 39.6% (TB) of the prequestions; the rest were left blank or answered “I don’t know”. In a beta-binomial model of the points earned on each prequestion, with random intercepts for participants and questions, prequestion scores were similar in the two conditions (model-estimated scores: 10.6% in the TB and 8.6% in the Polio condition), TB – Polio: 2.0 percentage points, 95% CrI [–2.0, 6.0], pd = 83.0%. Because each condition answered its own five

questions, this comparison cannot separate prior knowledge from the difficulty of the questions, so it is reported descriptively.

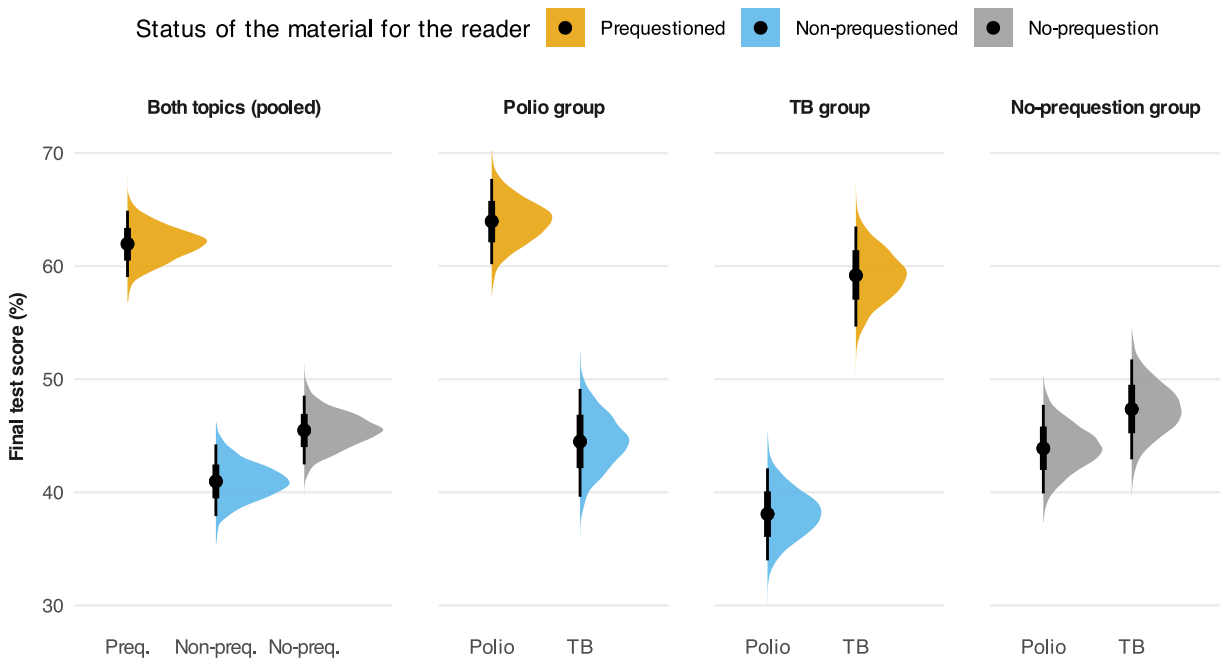
Final Test Performance

Final test performance was analysed as the proportion of each question's points earned (Figure 3). Pooled over the two prequestion conditions, participants earned 62.0% of the points on their prequestioned topic, compared with 45.5% for no-prequestion participants on the same questions, a specific benefit of 16.5 percentage points, 95% CrI [12.3, 20.6], pd > 99.9%, BF_{10} = 7.7×10^{5}: extreme evidence for a difference. On the non-prequestioned topic they earned 41.0%, slightly less than the no-prequestion condition, –4.5 percentage points, 95% CrI [–8.9, –0.2], pd = 98.0%, BF_{01} = 1.93. By the preregistered criterion (pd > 97.5%), this is a general effect, but a small cost rather than a benefit, and the Bayes factor was inconclusive, providing only anecdotal evidence for no difference.

In the Group × Topic model (Figure 3; Bayes factors on the response scale, see Statistical Analysis), the Polio group scored higher on Polio than on TB questions (63.9% vs. 44.5%), Polio – TB = 19.4 percentage points, 95% CrI [13.3, 25.4], pd > 99.9%, BF_{10} = 7.9×10^{4}, whereas the TB group scored higher on TB than on Polio questions (59.2% vs. 38.1%), Polio – TB = –21.1 percentage points, 95% CrI [–27.1, –15.1], pd > 99.9%, BF_{10} = 3.4×10^{6}. The no-prequestion group, which saw no prequestions, scored 43.9% on Polio and 47.3% on TB questions, Polio – TB = –3.4 percentage points, 95% CrI [–9.3, 2.5], pd = 87.4%, BF_{01} = 3.59. The Group × Topic interaction contrast—the Polio group's Polio – TB difference minus the TB group's—was 40.6 percentage points, 95% CrI [32.1, 49.1], pd > 99.9%, BF_{10} = 2.2×10^{11}: extreme evidence that each prequestion group did relatively better on its own topic.

Figure 3

Final Test Performance: Posterior Distributions of the Model-Estimated Means



Note. Posterior distributions of the model-estimated proportion of a question's points earned, averaged over the observed sample. Left: the status model, pooled over the two prequestion groups and both topics (prequestioned, non-prequestioned, and no-prequestion). Other panels: the Group × Topic model, for the Polio and TB questions in the Polio, TB, and no-prequestion groups; colours give the status of the questions for the readers in that group. Preq. = prequestioned; non-preq. = non-prequestioned; no-preq. = no-prequestion. Points are posterior means; thick and thin bars are 66% and 95% credible intervals.

Eye Movements

We first compare prequestioned and non-prequestioned sentences with the same sentences read by no-prequestion readers (Table 2), and then test the preregistered Group × Topic crossover (Figure 4). Figure 4 shows the posterior distributions of the means for every measure, pooled and by group.

Prequestioned and Non-Prequestioned Sentences Versus the No-Prequestion Condition

Prequestions changed how readers read the sentences that answered them, compared with no-prequestion readers reading the same sentences (Table 2). In the ex-Gaussian model of total gaze duration, prequestioned sentences had a longer tail of unusually long readings, τ: 33.9 ms per word, 95% CrI [11.9, 56.4], pd = 99.9%, BF_{10} = 31.07, very strong evidence for a difference. Typical reading time did not differ, μ: 2.8 ms per word, 95% CrI [−11.4, 17.0], pd = 64.7%, BF_{01} = 14.26, with strong evidence for no difference. Readers were more likely to return to prequestioned sentences from later text (37.6% vs. 30.2%), 7.3 percentage points, 95% CrI [2.4, 12.2], pd = 99.8%, BF_{10} = 1.67, anecdotal evidence. Prequestioned sentences also received more re-reading: the expected re-reading time per sentence, including sentences that were not re-read, was 128.9 ms per word compared with 102.2 ms per word, 26.7 ms per word, 95% CrI [8.0, 45.9], pd = 99.7%, BF_{10} = 2.35, anecdotal evidence. In the two parts of the hurdle model, this reflected re-reading more often—prequestioned sentences were 9.5 percentage points more likely to be re-read (pd > 99.9%, BF_{10} = 44.97)—rather than longer re-reading once a sentence was re-read (19.6 ms per word, 95% CrI [−6.0, 45.3], pd = 93.5%, BF_{01} = 2.57). First-pass duration was not reliably longer, 12.1 ms per word, 95% CrI [−7.2, 31.6], pd = 88.9%, BF_{01} = 4.45; the Bayes factor favoured no difference, but with only moderate evidence.

Non-prequestioned sentences, by contrast, were read like the same sentences in the no-prequestion condition. No measure showed a general effect (non-prequestioned − no-prequestion, Table 2; all pd < 97.5%), and on every measure the Bayes factor favoured its absence (BF_{01} from 1.9 to 15.4), with moderate or stronger evidence for 4 of the 5 measures and anecdotal evidence for the rest. In the hurdle part of the re-reading model, non-prequestioned sentences were also somewhat more likely to be re-read than without prequestions (6.1 percentage points, pd = 98.3%, BF_{10} = 1.73), but the expected re-reading time did not differ. The effects of prequestions on reading were therefore largely specific to the prequestioned sentences.

Table 2

Prequestioned and Non-Prequestioned Sentences Versus the Same Sentences Read Without Prequestions

Measure	Prequestioned – No-prequestion	pd	BF_{10}	Non-prequestioned – No-prequestion	pd	BF_{01}
Total gaze: μ	2.8 [–11.4, 17.0]	64.7%	0.07	2.3 [–11.2, 15.9]	63.1%	15.41
Total gaze: τ	33.9 [11.9, 56.4]	99.9%	31.07	–4.7 [–24.0, 14.7]	68.5%	3.72
Re-reading duration	26.7 [8.0, 45.9]	99.7%	2.35	10.8 [–7.2, 28.9]	88.0%	1.90
Second-pass regression	7.3 [2.4, 12.2]	99.8%	1.67	–1.3 [–6.1, 3.5]	70.6%	6.12
First-pass duration	12.1 [–7.2, 31.6]	88.9%	0.22	–8.8 [–26.9, 9.3]	83.3%	6.00

Note. Posterior means and 95% credible intervals of the differences, pooled over the Polio and TB prequestion conditions and averaged over the two topics, in ms per word (durations) or percentage points (second-pass regressions). Prequestioned – no-prequestion is the specific effect and non-prequestioned – no-prequestion the general effect of prequestions. Total gaze duration was analysed with an ex-Gaussian model (μ = typical reading time; τ = tail of unusually long readings); re-reading duration with a hurdle Gamma model (expected re-reading time per sentence, including sentences not re-read as 0); first-pass duration with a Gamma model; second-pass regressions with a logistic model. pd = probability of direction. BF_{10} = Bayes factor for a difference (Savage–Dickey); $BF_{01} = 1/BF_{10}$, evidence for no difference. First-pass duration is a local measure; all others are global.

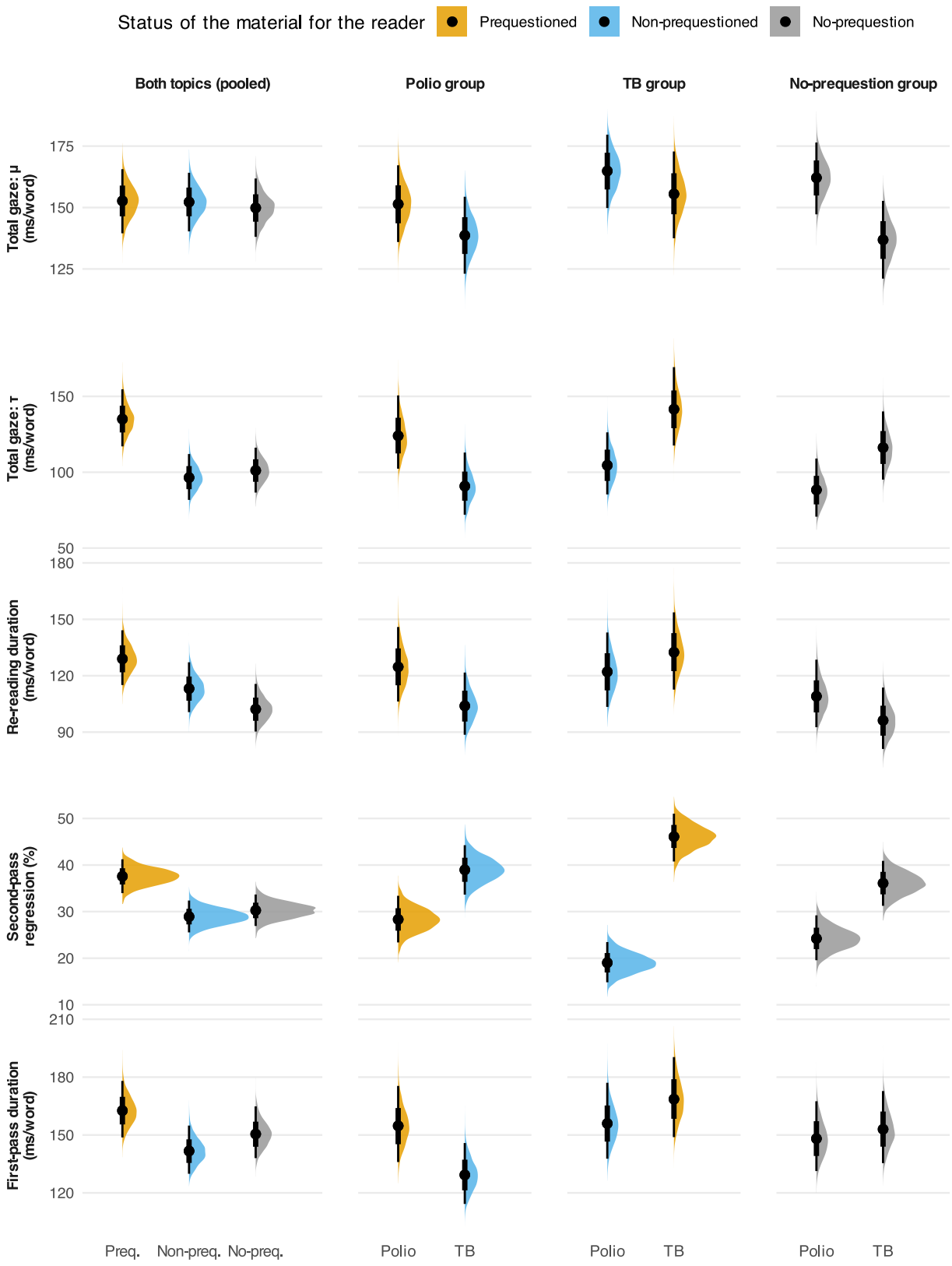
Group × Topic

The preregistered prediction was a crossover: longer reading of Polio than of TB sentences in the Polio group, the reverse in the TB group, and no difference in the no-prequestion

group, so that the Group × Topic interaction contrast would be positive. The interaction contrast was positive (pd > 95%) for the tail of unusually long readings (τ): 70.1 ms per word, 95% CrI [29.2, 112.0], pd = 99.9%, BF_{10} = 109; look-backs from later text: 16.5 percentage points, 95% CrI [6.9, 26.0], pd > 99.9%, BF_{10} = 63.06; and first-pass reading: 37.9 ms per word, 95% CrI [1.1, 75.0], pd = 97.8%, BF_{10} = 1.50. It was not reliably positive for typical reading time (μ): 3.5 ms per word, 95% CrI [–23.4, 30.5], pd = 60.1%, BF_{01} = 9.06 and re-reading time: 30.5 ms per word, 95% CrI [–7.3, 68.5], pd = 94.2%, BF_{10} = 1.00. For the tail of total gaze duration (τ), the Polio – TB difference was 33.2 ms per word, 95% CrI [4.6, 62.7], pd = 98.9%, BF_{10} = 4.24 in the Polio group, –36.9 ms per word, 95% CrI [–68.6, –6.1], pd = 99.0%, BF_{10} = 5.07 in the TB group, and –27.9 ms per word, 95% CrI [–55.9, –1.0], pd = 97.9%, BF_{10} = 2.27 in the no-prequestion group. For look-backs from later text (Bayes factors on the response scale), it was –10.6 percentage points, 95% CrI [–17.7, –3.4], pd = 99.8%, BF_{10} = 10.73 in the Polio group, –27.2 percentage points, 95% CrI [–33.7, –20.5], pd > 99.9%, $BF_{10} = 2.2 \times 10^{7}$ in the TB group, and –11.9 percentage points, 95% CrI [–18.6, –5.1], pd > 99.9%, BF_{10} = 58.69 in the no-prequestion group. Because the Polio and TB sentences differ, the no-prequestion group shows how the topics are read without prequestions (Figure 4).

Figure 4

Eye-Movement Measures: Posterior Distributions of the Model-Estimated Means



Note. Posterior distributions of the model-estimated means, averaged over the observed sample. Left: the status models, pooled over the two prequestion groups and both topics (prequestioned, non-prequestioned, and no-prequestion sentences). Other panels: the Group × Topic models, for the Polio and TB sentences in the Polio, TB, and no-prequestion groups; colours give the status of the sentences for the readers in that group. μ and τ are the Gaussian mean (typical reading time) and the tail mean (unusually long readings) of the ex-Gaussian model of total gaze duration. Durations are in ms per word; second-pass regressions are the percentage of sentences returned to from later text. Preq. = prequestioned; non-preq. = non-prequestioned; no-preq. = no-prequestion. Points are posterior means; thick and thin bars are 66% and 95% credible intervals.

Discussion

This study used eye tracking to test whether prequestions improve memory by directing attention toward the information they address. Two sets of preregistered comparisons addressed this question. Pooling the two prequestion groups, we compared each target sentence and question when it answered the reader's prequestions with the same sentence or question for readers who saw no prequestions (the specific effect), and the other topic's material with the same material without prequestions (the general effect). The Group × Topic comparisons then asked whether the effects reversed with the prequestioned topic, that is, whether the Polio group favoured Polio material and the TB group TB material. On the final test, prequestions produced the typical specific benefit and a small general cost on the non-prequestioned topic, for which the Bayes factor was inconclusive, and the crossover was clear: each prequestion group did relatively better on its own topic (interaction contrast, extreme evidence). Eye movements showed the same specificity. Compared with the same sentences read without prequestions, prequestioned sentences received more unusually long readings, were re-read more often, and were more likely to be looked back to from later text, whereas non-prequestioned sentences were read like the same sentences in the no-prequestion condition on every measure. In the Group ×

Topic comparisons, the interaction contrast was positive (pd > 95%) for the tail of unusually long readings (τ), look-backs from later text, and first-pass reading; and not reliably positive for typical reading time (μ) and re-reading time (Figure 4). Prequestions therefore changed how readers treated the information they targeted, and only that information.

The pattern of eye-movement effects indicates what kind of attention prequestions recruit. First, prequestions did not make readers read the relevant material more slowly overall: the Bayes factors favoured no difference from the no-prequestion condition in typical reading time (μ) and, more weakly, in first-pass duration. Prequestions therefore do not seem to act as a general instruction to read prequestion-relevant content with more care. Second, the additional processing appeared in a subset of encounters—as occasional unusually long readings in the tail of the reading-time distribution (τ)—and in returns to the sentences: readers re-read prequestioned sentences more often and looked back to them more often from later text, although re-reading was not reliably longer once it occurred. Third, the additional processing was confined to the prequestioned sentences: the non-prequestioned sentences, interleaved with them in the same text, were read as they were without prequestions, except that they were somewhat more likely to be re-read.

Together, these features suggest that additional processing is triggered when a reader recognizes that a sentence answers one of the prequestions. Recognizing an answer requires matching the sentence against the prequestions held in memory, so it can occur only once enough of the sentence has been read, and it can fail—a reader may not notice the connection, or may no longer remember the question. Recognition may also be what makes prequestions effective: in video lectures, prequestions enhanced learning only when learners remembered them (St. Hilaire & Carpenter, 2020). Recognition would therefore add processing to some encounters but not others, lengthening the tail of the reading-time distribution rather than shifting the whole distribution, as the effect on τ but not μ shows. Ex-Gaussian analyses of fixation durations in

reading have shown that the two components can be affected separately. Word frequency affects both μ and τ (Staub et al., 2010), whereas a modest degradation of the text, such as presenting it faintly, increases only μ, slowing all fixations by a similar amount without adding unusually long ones (White & Staub, 2012). Prequestions showed the reverse pattern: they did not shift typical reading time but lengthened only the tail. They therefore did not make the prequestioned sentences harder or slower to read in general; instead, they added processing to a subset of encounters with them. Memory-based processes have been found to act on τ in this way in scene viewing, where familiarity with an image lengthened only a subset of relatively long fixations (Guy et al., 2020). Once an answer is recognized, readers may pause to compare it with the question or with the guess they gave before reading and to encode it in relation to the question, and they may return to it later to check or rehearse it, as the increased re-reading and look-backs suggest. Part of this pattern fits the attentional-window account (Sana & Carpenter, 2023), on which prequestions heighten attention while learners search for the answers, until the answers are found: the long readings of the answer sentences could mark the point at which a reader finds an answer. The account does not, however, predict the look-backs. Once an answer has been found, attention should return to its usual level, yet readers returned to prequestioned sentences from later text more often than no-prequestion readers returned to the same sentences. Prequestions therefore appear to engage readers with an answer beyond the moment it is found: after recognizing it, readers return to it, perhaps to check it against the question or against their earlier guess, or to rehearse it. The recognition account covers both features of the data—pausing when an answer is recognized and returning to it later—whereas the attentional-window account covers only the first.

The recognition account is also what the mental model hypothesis predicts (Motz et al., 2026). On that hypothesis, prequestions activate relevant knowledge, information related to it resonates with that knowledge when it is encountered, and additional attention follows

recognition rather than preceding it. Consistent with this sequence, prequestions here did not lengthen first-pass reading or typical reading time of the prequestioned sentences, as heightened attention from the start would; instead, they added processing once an answer had been read and drew readers back to it later. The present design, however, tests only part of the hypothesis. The non-prequestioned sentences were about the other disease, so they do not show whether activated knowledge also facilitates the processing of related, non-prequestioned information on the prequestioned topic.

The pattern also adds to evidence that attention is not the whole story. If prequestions worked simply by heightening attention to the content they address, readers should have read that content more slowly or carefully throughout, shifting the whole distribution of reading times, that is, μ. Instead, typical reading time did not change—the Bayes factor favoured no difference in μ—and prequestions lengthened only the tail of the distribution (τ) and increased returns to the answers. Together with findings that prequestions can improve learning without increasing the time spent on the material (Motz et al., 2026) and that pretesting does not always reduce mind-wandering (Welhaf et al., 2022), this suggests that heightened attention alone does not explain prequestion effects. The occasional long pauses and the returns to the answers instead resemble metacognitive monitoring. On a metacognitive account, answering prequestions shows learners what they do not know, and they then monitor the material for the information that fills these gaps (King-Shepard et al., 2025): noticing that a sentence answers a question, comparing it with the guess given before reading, and checking it later. Recognizing an answer is itself a metacognitive judgment, so the recognition-triggered processing proposed here can be understood as metacognitive monitoring expressed in eye movements. Because monitoring is itself expressed through where and for how long readers look, eye movements alone cannot separate the two accounts. Combining eye tracking with direct measures of metacognition, such

as judgments of learning or confidence in each answer, as recently recommended (King-Shepard et al., 2025), could do so.

The specificity of the effects also constrains the mechanism. A specific benefit on its own is consistent with any of the proposed mechanisms—attention, elaboration, curiosity, or metacognition—but the absence of a general benefit is not consistent with accounts on which prequestions raise curiosity or prompt metacognitive monitoring for the topic as a whole (King-Shepard et al., 2025). In their meta-analysis, King-Shepard et al. (2025) found no general benefit, and also no general cost, which suggests that if prequestions orient attention toward the prequestioned information, this orienting does not impede learning of other information. The eye movements here fit this view: non-prequestioned sentences were read as they were without prequestions on every measure, so prequestions changed how readers treated their answers, not how they read the topic or the passage as a whole. The cost on the non-prequestioned topic in the final test is therefore unlikely to reflect attention diverted from that material during reading; it may instead arise at encoding or retrieval, for example from interference by the prequestioned topic.

Whether these episodes of additional processing produce the memory benefit cannot be determined from these data. Reading was observed rather than manipulated, and the eye-movement effects were modest compared with the memory benefit, so prequestions may also strengthen memory through processes that leave little trace in reading time, such as how an answer is encoded once it is recognized or how it is related to the earlier retrieval attempt. This possibility is consistent with evidence that prequestions can improve learning without increasing the time learners spend on the material (Motz et al., 2026). Experiments that manipulate whether and when readers can return to the answers, or that record recognition of answers directly, could test whether recognition-triggered processing is necessary for the benefit.

These findings agree with the two previous eye-tracking studies of prequestions in showing that prequestions shift attention toward the information they address. In Lewis and

Mensink (2012), TB prequestions increased first-pass reading of the targeted sentences by increasing re-reading within those sentences before readers moved on, rather than by slowing forward reading, and increased the time spent looking back to them after moving on. Readers in both studies were more likely to look back to prequestioned sentences from later in the text. Both studies also found more time spent re-reading prequestioned sentences, although here this reflected re-reading them more often rather than for longer. The studies differ in first-pass reading, which was not reliably longer than in the no-prequestion condition here. Earlier studies also analysed average reading and fixation times, whereas the present analysis separated typical reading time (μ) from the tail of unusually long readings (τ). This distinction matters for prequestions: the extra processing here was confined to a subset of encounters, which an analysis of means would dilute and could not distinguish from slower reading overall. Several differences between the studies may contribute. Lewis and Mensink (2012) compared a single prequestion group with a control group of about 21 readers each, whereas the present design pooled two prequestion groups that targeted different sentences, so that every target sentence was compared with and without prequestions in a larger sample. Re-reading is also defined differently across studies and eye-tracking software; for example, re-reading duration is sometimes computed as go-past time minus first-pass time, which mostly reflects time spent on earlier text, whereas we measured re-reading as time spent on the target sentence itself after first-pass reading. Finally, we corrected for the eyes landing partway down a page when it first appears, which otherwise misclassifies the first reading of sentences near the top of a page as re-reading. Bostan and Ozcelik (2026) likewise found more fixations and longer fixation time on prequestioned sentences, and more transitions between prequestioned sentences, which parallels the look-backs we observed. Unlike the present results and most prequestion research, they also reported a general benefit. Their integrative prequestions required combining information from different parts of the text, which may have directed attention to non-prequestioned information as well,

consistent with the view that general benefits arise when prequestions direct attention beyond the tested information (Pan & Carpenter, 2023). In the present study, prequestions about single facts neither increased attention to non-prequestioned information nor improved memory for it. Memory for the non-prequestioned material was, if anything, slightly worse than in the no-prequestion condition (-4.5 percentage points, 95% CrI [–8.9, –0.2], pd = 98.0%, BF_{01} = 1.93). This difference was not reliable, and the Bayes factor did not favour it, but it is consistent with a trade-off in which prequestions draw attention and encoding toward the prequestioned topic at the expense of the rest of the text. Testing this possibility would require more questions per topic. With 12 participants per group, however, the estimates of Bostan and Ozcelik (2026) are imprecise.

Several limitations should be noted. Each topic was represented by only five target sentences and five test questions. The within-group Polio – TB differences therefore combine any effect of prequestions with differences between the materials: the Polio and TB sentences differed in length, content, reading level, and position on the page, and the no-prequestion group read them differently even without prequestions. The Group × Topic interaction contrasts remove these differences, but they compare two groups of readers and rest on five items per topic. We therefore based our conclusions primarily on the pooled comparisons, which compare the same sentences and questions with and without prequestions, so that differences between the materials cancel out, and we treat the Group × Topic contrasts as converging evidence where they were reliable. Texts with more target sentences per topic would allow topic-specific effects to be estimated precisely. The comparisons with the no-prequestion condition are between participants, and so are less precise than comparisons within readers. The Bayes factors depend on the priors on the effects: wider priors favour the absence of an effect more strongly, and narrower priors less so. They show that the data are better described without a general effect than with an effect of the size the priors allow, not that a small general effect is impossible. Eye movements were

recorded without head stabilization, and recorded gaze positions drifted vertically for some readers. The measures were therefore derived from the fixation data with a drift correction bounded by the calibration accuracy criterion; drift beyond that bound was not corrected. Finally, responses were scored with an automated rubric, and the sample came from a single university.

In sum, prequestions guide where readers look, and only toward the information they target: rather than reading the relevant material more slowly, readers pause on some of the answers and return to them later, while reading the rest of the text as they would without prequestions. Prequestions therefore do not simply heighten attention to the material they address. Such recognition-triggered processing is a plausible route by which prequestions strengthen memory for their answers, but whether it accounts for the memory benefit remains to be tested.

References

Arel-Bundock, V., Greifer, N., & Heiss, A. (2024). How to interpret statistical models using marginaleffects for R and Python. *Journal of Statistical Software*, *111*(9), 1–32. https://doi.org/10.18637/jss.v111.i09

Balota, D. A., & Yap, M. J. (2011). Moving beyond the mean in studies of mental chronometry: The power of response time distributional analyses. *Current Directions in Psychological Science*, *20*(3), 160–166. https://doi.org/10.1177/0963721411408885

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255–278. https://doi.org/10.1016/j.jml.2012.11.001

Benoit, K., Watanabe, K., Wang, H., Lua, J. W., & Kuha, J. (2024). *Quanteda.textstats: Textual statistics for the quantitative analysis of textual data*. https://quanteda.io

Benoit, K., Watanabe, K., Wang, H., Nulty, P., Obeng, A., Müller, S., & Matsuo, A. (2018). Quanteda: An r package for the quantitative analysis of textual data. *Journal of Open Source Software*, *3*(30), 774. https://doi.org/10.21105/joss.00774

Bjork, R. A. (1994). Memory and metamemory considerations in the training of human beings. In J. Metcalfe & A. P. Shimamura (Eds.), *Metacognition: Knowing about knowing* (pp. 185–205). MIT Press.

Bostan, E., & Ozcelik, E. (2026). The effect of integrative prequestions on learning from text: An eye-tracking study. *The Journal of Experimental Education*, *94*(1), 33–45. https://doi.org/10.1080/00220973.2024.2355980

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, *80*(1), 1–28. https://doi.org/10.18637/jss.v080.i01

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*(1), 1–32. https://doi.org/10.18637/jss.v076.i01

Carpenter, S. K. (2012). Testing enhances the transfer of learning. *Current Directions in Psychological Science*, *21*(5), 279–283. https://doi.org/10.1177/0963721412452728

Carpenter, S. K., Rahman, S., & Perkins, K. (2018). The effects of prequestions on classroom learning. *Journal of Experimental Psychology: Applied*, *24*(1), 34–42. https://doi.org/10.1037/xap0000145

Carpenter, S. K., & Toftness, A. R. (2017). The effect of prequestions on learning from video presentations. *Journal of Applied Research in Memory and Cognition*, *6*(1), 104–109. https://doi.org/10.1016/j.jarmac.2016.07.014

Carr, J. W., Pescuma, V. N., Furlan, M., Ktori, M., & Crepaldi, D. (2022). Algorithms for the automated correction of vertical drift in eye-tracking data. *Behavior Research Methods*, *54*(1), 287–310. https://doi.org/10.3758/s13428-021-01554-0

Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest*, *14*(1), 4–58. https://doi.org/10.1177/1529100612453266

Firke, S. (2024). *Janitor: Simple tools for examining and cleaning dirty data*. https://github.com/sfirke/janitor

Gabry, J., Češnovar, R., Johnson, A., & Bronder, S. (2025). *Cmdstanr: R interface to CmdStan*. https://mc-stan.org/cmdstanr/

Geller, J., Carpenter, S. K., Lamm, M. H., Rahman, S., Armstrong, P. I., & Coffman, C. R. (2017). Prequestions do not enhance the benefits of retrieval in a STEM classroom.

Cognitive Research: Principles and Implications, *2*(1), 42. https://doi.org/10.1186/s41235-017-0078-z

Geller, J., Kubinec, R., Parlett Pelleriti, C. M., & Vuorre, M. (in press). A beta way: A tutorial on Bayesian beta regression for psychological research. *Behavior Research Methods*. https://doi.org/10.31234/osf.io/d6v5t_v3

Guy, N., Lancry-Dayan, O. C., & Pertzov, Y. (2020). Not all fixations are created equal: The benefits of using ex-Gaussian modeling of fixation durations. *Journal of Vision*, *20*(10), 9. https://doi.org/10.1167/jov.20.10.9

Just, M. A., & Carpenter, P. A. (1980). A theory of reading: From eye fixations to comprehension. *Psychological Review*, *87*(4), 329–354. https://doi.org/10.1037/0033-295X.87.4.329

Kallioinen, N., Paananen, T., Bürkner, P.-C., & Vehtari, A. (2023). Detecting and diagnosing prior and likelihood sensitivity with power-scaling. *Statistics and Computing*, *34*. https://doi.org/10.1007/s11222-023-10366-5

Kay, M. (2024). ggdist: Visualizations of distributions and uncertainty in the grammar of graphics. *IEEE Transactions on Visualization and Computer Graphics*, *30*(1), 414–424. https://doi.org/10.1109/TVCG.2023.3327195

Kincaid, J. P., Fishburne, R. P., Rogers, R. L., & Chissom, B. S. (1975). *Derivation of new readability formulas (automated readability index, fog count and Flesch reading ease formula) for Navy enlisted personnel* (Nos. Research Branch Report 8-75). Naval Technical Training Command.

King-Shepard, Q. W., Walker, J., Nokes-Malach, T. J., Carpenter, S. K., & Fraundorf, S. H. (2025). The effect of prequestions on learning: A multilevel meta-analysis. *Educational Psychology Review*, *37*, 115. https://doi.org/10.1007/s10648-025-10075-7

Koo, T. K., & Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, *15*(2), 155–163. https://doi.org/10.1016/j.jcm.2016.02.012

Kornell, N., & Vaughn, K. E. (2016). How retrieval attempts affect learning: A review and synthesis. In B. H. Ross (Ed.), *Psychology of learning and motivation* (Vol. 65, pp. 183–215). Academic Press. https://doi.org/10.1016/bs.plm.2016.03.003

Kubinec, R. (2023). Ordered beta regression: A parsimonious, well-fitting model for continuous data with lower and upper bounds. *Political Analysis*, *31*(4), 519–536. https://doi.org/10.1017/pan.2022.20

Kubinec, R. (2025). *Ordbetareg: Ordered beta regression models with brms*. https://doi.org/10.32614/CRAN.package.ordbetareg

Kumle, L., Võ, M. L.-H., & Draschkow, D. (2021). Estimating power in (generalized) linear mixed models: An open introduction and tutorial in R. *Behavior Research Methods*, *53*(6), 2528–2543. https://doi.org/10.3758/s13428-021-01546-0

Lee, M. D., & Wagenmakers, E.-J. (2014). *Bayesian cognitive modeling: A practical course*. Cambridge University Press. https://doi.org/10.1017/CBO9781139087759

Lewis, M. R., & Mensink, M. C. (2012). Prereading questions and online text comprehension. *Discourse Processes*, *49*(5), 367–390. https://doi.org/10.1080/0163853X.2012.662801

Liu, H. (2008). Dependency distance as a metric of language comprehension difficulty. *Journal of Cognitive Science*, *9*(2), 159–191.

Lo, S., & Andrews, S. (2015). To transform or not to transform: Using generalized linear mixed models to analyse reaction time data. *Frontiers in Psychology*, *6*, 1171. https://doi.org/10.3389/fpsyg.2015.01171

Lüdecke, D., Waggoner, P., & Makowski, D. (2019). insight: A unified interface to access information from model objects in R. *Journal of Open Source Software*, *4*(38), 1412. https://doi.org/10.21105/joss.01412

Makowski, D. (2026). *Cogmod: Cognitive models for subjective scales and decision making tasks*. https://github.com/DominiqueMakowski/cogmod

Makowski, D., Ben-Shachar, M. S., Chen, S. H. A., & Lüdecke, D. (2019). Indices of effect existence and significance in the Bayesian framework. *Frontiers in Psychology*, *10*, 2767. https://doi.org/10.3389/fpsyg.2019.02767

Makowski, D., Ben-Shachar, M. S., & Lüdecke, D. (2019). bayestestR: Describing effects and their uncertainty, existence and significance within the bayesian framework. *Journal of Open Source Software*, *4*(40), 1541. https://doi.org/10.21105/joss.01541

Makowski, D., Ben-Shachar, M. S., Wiernik, B. M., Patil, I., Thériault, R., & Lüdecke, D. (2025). modelbased: An R package to make the most out of your statistical models through marginal means, marginal effects, and model predictions. *Journal of Open Source Software*, *10*(109), 7969. https://doi.org/10.21105/joss.07969

Matzke, D., & Wagenmakers, E.-J. (2009). Psychological interpretation of the ex-Gaussian and shifted Wald parameters: A diffusion model analysis. *Psychonomic Bulletin & Review*, *16*(5), 798–817. https://doi.org/10.3758/PBR.16.5.798

Motz, B. A., Chinni, A., de Leeuw, J. R., Jankowski, H. E., Aggarwal, A., Amato, M., Berlin, K., Britten, K., Brown, A., Cerchiaro, M. L., Evans, N. J., Findley, A., Gorman, R., Gregg, K. D., Hansen, K., Hollender, H., Hullinger, R. A., Larkin, P., Lion, M., … Fyfe, E. R. (2026). ManyClasses 2: The effects of prequestions on media interactions and learning. *Journal of Educational Psychology*, *118*(2), 153–175. https://doi.org/10.1037/edu0000978

Olsen, A. (2012). *The Tobii I-VT fixation filter: Algorithm description*. Tobii Technology.

Pan, S. C., & Carpenter, S. K. (2023). Prequestioning and pretesting effects: A review of empirical research, theoretical perspectives, and implications for educational practice. *Educational Psychology Review*, *35*, 97. https://doi.org/10.1007/s10648-023-09814-5

Pan, S. C., Sana, F., Schmitt, A. G., & Bjork, E. L. (2020). Pretesting reduces mind wandering and enhances learning during online lectures. *Journal of Applied Research in Memory and Cognition*, *9*(4), 542–554. https://doi.org/10.1016/j.jarmac.2020.07.004

Pedersen, T. L. (2025). *Patchwork: The composer of plots*. https://patchwork.data-imaginist.com

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://doi.org/10.32614/R.manuals

Rayner, K. (1998). Eye movements in reading and information processing: 20 years of research. *Psychological Bulletin*, *124*(3), 372–422. https://doi.org/10.1037/0033-2909.124.3.372

Richland, L. E., Kornell, N., & Kao, L. S. (2009). The pretesting effect: Do unsuccessful retrieval attempts enhance learning? *Journal of Experimental Psychology: Applied*, *15*(3), 243–257. https://doi.org/10.1037/a0016496

Roediger, H. L., & Karpicke, J. D. (2006). The power of testing memory: Basic research and implications for educational practice. *Perspectives on Psychological Science*, *1*(3), 181–210. https://doi.org/10.1111/j.1745-6916.2006.00012.x

Rothkopf, E. Z., & Billington, M. J. (1979). Goal-guided learning from text: Inferring a descriptive processing model from inspection times and eye movements. *Journal of Educational Psychology*, *71*(3), 310–327. https://doi.org/10.1037/0022-0663.71.3.310

Rowland, C. A. (2014). The effect of testing versus restudy on retention: A meta-analytic review of the testing effect. *Psychological Bulletin*, *140*(6), 1432–1463. https://doi.org/10.1037/a0037559

Salvucci, D. D., & Goldberg, J. H. (2000). Identifying fixations and saccades in eye-tracking protocols. *Proceedings of the 2000 Symposium on Eye Tracking Research & Applications*, 71–78. https://doi.org/10.1145/355017.355028

Sana, F., & Carpenter, S. K. (2023). Broader benefits of the pretesting effect: Placement matters. *Psychonomic Bulletin & Review*. https://doi.org/10.3758/s13423-023-02274-6

Schotter, E. R., & Dillon, B. (2025). A beginner's guide to eye tracking for psycholinguistic studies of reading. *Behavior Research Methods*, *57*(2), 68. https://doi.org/10.3758/s13428-024-02572-4

Serrano-Carot, M., Angele, B., Xu, H., & Vasilev, M. R. (2025). *Webcams can be used to study eye movements during reading*. PsyArXiv preprint; PsyArXiv. https://doi.org/10.31234/osf.io/bzt2h_v1

Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, *86*(2), 420–428. https://doi.org/10.1037/0033-2909.86.2.420

Silveira, N., Dozat, T., Marneffe, M.-C. de, Bowman, S., Connor, M., Bauer, J., & Manning, C. (2014). A gold standard dependency corpus for English. *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*, 2897–2904.

St. Hilaire, K. J., & Carpenter, S. K. (2020). Prequestions enhance learning, but only when they are remembered. *Journal of Experimental Psychology: Applied*, *26*(4), 705–716. https://doi.org/10.1037/xap0000296

St. Hilaire, K. J., Carpenter, S. K., & Jennings, J. M. (2019). Using prequestions to enhance learning from reading passages: The roles of question type and structure building ability. *Memory*, *27*(9), 1204–1213. https://doi.org/10.1080/09658211.2019.1641209

St. Hilaire, K. J., Chan, J. C. K., & Ahn, D. (2024). Guessing as a learning intervention: A meta-analytic review of the prequestion effect. *Psychonomic Bulletin & Review*, *31*(2), 411–441. https://doi.org/10.3758/s13423-023-02353-8

Staub, A., White, S. J., Drieghe, D., Hollway, E. C., & Rayner, K. (2010). Distributional effects of word frequency on eye fixation durations. *Journal of Experimental Psychology: Human Perception and Performance*, *36*(5), 1280–1293. https://doi.org/10.1037/a0016896

Straka, M., & Straková, J. (2017). Tokenizing, POS tagging, lemmatizing and parsing UD 2.0 with UDPipe. *Proceedings of the CoNLL 2017 Shared Task: Multilingual Parsing from Raw Text to Universal Dependencies*, 88–99. https://doi.org/10.18653/v1/K17-3009

Urbanek, S. (2026). *Png: Read and write PNG images*. https://www.rforge.net/png/

Wagenmakers, E.-J., Lodewyckx, T., Kuriyal, H., & Grasman, R. (2010). Bayesian hypothesis testing for psychologists: A tutorial on the Savage–Dickey method. *Cognitive Psychology*, *60*(3), 158–189. https://doi.org/10.1016/j.cogpsych.2009.12.001

Welhaf, M. S., Phillips, N. E., Smeekens, B. A., Miyake, A., & Kane, M. J. (2022). Interpolated testing and content pretesting as interventions to reduce task-unrelated thoughts during a video lecture. *Cognitive Research: Principles and Implications*, *7*, 26. https://doi.org/10.1186/s41235-022-00372-y

White, S. J., & Staub, A. (2012). The distribution of fixation durations during reading: Effects of stimulus quality. *Journal of Experimental Psychology: Human Perception and Performance*, *38*(3), 603–617. https://doi.org/10.1037/a0025338

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., … Yutani, H. (2019).

Welcome to the tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686

Wickham, H., Pedersen, T. L., & Seidel, D. (2025). *Scales: Scale functions for visualization*. https://scales.r-lib.org

Wijffels, J., & Straka, M. (2026). *Udpipe: Tokenization, parts of speech tagging, lemmatization and dependency parsing with the UDPipe 'NLP' toolkit*. https://bnosac.github.io/udpipe/en/index.html

Xie, Y. (2015). *Dynamic documents with R and knitr* (2nd ed.). Chapman; Hall/CRC. https://yihui.org/knitr/

Xie, Y. (2026). *Xfun: Supporting functions for packages maintained by yihui xie*. https://github.com/yihui/xfun

Yang, J., Zhang, Y., Pi, Z., & Xie, Y. (2021). Students' achievement motivation moderates the effects of interpolated pre-questions on attention and learning from video lectures. *Learning and Individual Differences*, *91*, 102055. https://doi.org/10.1016/j.lindif.2021.102055

Appendix A

Questions and Scoring Rules

Prequestion and final test responses were scored with the same rule-based script (`scoring_rubric.R`), written by Claude Opus 5.5 (Anthropic; model ID `claude-opus-5-5`) and applied identically to every response. Each question awarded one point per correct element present in the response (Table A1, Table A2). Blank responses, "I don't know" and similar responses, and random keystrokes scored 0. Minor misspellings were accepted when the intended word was unambiguous (e.g., "afganistan"). The same rules were used for the prequestions and the final test.

Table A1

Final Test Questions and Answers

Q	Topic	Question	Answer	Points
1	Polio	What percent of the global budget does the Gates Foundation devote toward the eradication of polio	17%	1
2	Polio	Is there a cure for polio?	no	1
3	Polio	In which countries is polio still a major concern?	Nigeria, India, Pakistan, and Afghanistan	4
4	Polio	What measures are included in the modern treatment of polio?	antibiotics to prevent infections in weakened muscles, analgesics for pain, moderate exercise, and a nutritious diet	4
5	TB	What are the symptoms of tuberculosis?	a chronic cough with blood-tinged sputum, fever, night sweats, and weight loss	4
6	TB	What percentage of new infections will turn into active cases of tuberculosis?	10%	1
7	TB	What is the treatment for tuberculosis?	a long-term course of two antibiotics to kill the bacteria.	3
8	TB	Can people with latent tuberculosis spread the disease?	no	1
9	TB	How is tuberculosis spread through the air?	cough, sneeze, or spit	3
10	Polio	Since the development of the polio vaccine, the global number of polio cases per year has dropped from _____ to _____.	many hundreds of thousands to around a thousand	2

Note. Points = maximum points for the question. The Polio prequestion condition answered questions 1–4 and 10 before reading; the TB prequestion condition answered questions 5–9.

Table A2

Scoring Rules and Agreement With the Second Rater for Each Question

Q	Credited elements	*n*	Exact	ICC
1	The number 17 (as a whole number, e.g., not 170).	156	100.0%	1.00
2	A negative answer: “no”, “no cure”, or “not a cure”.	196	100.0%	1.00
3	One point each for Nigeria, India, Pakistan, and Afghanistan.	163	96.9%	.99
4	One point each for antibiotics; analgesics or pain relief; exercise (including physical activity or therapy); and diet or nutrition. “Antibiotic resistance” alone was not credited.	160	98.8%	.99
5	One point each for cough (or blood-tinged sputum/phlegm); fever (or temperature); night sweats; and weight loss.	179	91.6%	.96
6	10% (or “ten percent”, 1/10); not credited if stated as “less than” or “under” 10%.	155	99.4%	.99
7	One point each for antibiotics (or antibacterials); two (two drugs, courses, rounds, or types); and long-term (long, extended, prolonged, or months).	167	97.6%	.97
8	A negative answer (“no”).	184	100.0%	1.00
9	One point each for cough, sneeze, and spit (or saliva).	173	99.4%	1.00
10	One point each for a starting value of hundreds of thousands (100,000–999,999) and an end value of about a thousand (500–9,999, or “a thousand”).	132	90.9%	.94

Note. One point was awarded for each credited element present in a response, up to the question’s maximum (see Table A1). *n* = number of non-blank responses (prequestions and final

test) scored by both the automated rubric and the second rater; Exact = percentage of responses given the same score; ICC = ICC(2,1), absolute agreement.

Appendix B

Model Specifications

Table B1

Bayesian Models: Families, Effects, and Priors

Measure	Family	Random effects	Prior on comparisons
Total gaze duration	Ex-Gaussian (μ identity; σ, τ softplus)	μ and τ: participants (intercept, Topic slope), sentences (intercept, Status or Group slope); σ: participants (intercept)	μ: normal(0, 0.1) s/word; τ: normal(0, 0.5)
First-pass duration	Gamma (log)	Participants (intercept, Topic slope), sentences (intercept, Status or Group slope)	normal(0, 0.5)
Re-reading duration	Hurdle Gamma (Gamma: log; hurdle: logit)	Both parts: participants (intercept, Topic slope), sentences (intercept, Status or Group slope)	Gamma: normal(0, 0.5); hurdle: normal(0, 1.5)
Second-pass regression	Bernoulli (logit)	Participants (intercept, Topic slope), sentences (intercept, Status or Group slope)	normal(0, 1.5)
Final test	Beta-binomial (logit)	Participants (intercept, Topic slope), questions (intercept, Status or Group slope)	normal(0, 1.5)
Prequestions	Beta-binomial (logit)	Participants (intercept), questions (intercept)	normal(0, 1.5)

Note. Pooled models: Status (prequestioned, non-prequestioned, no-prequestion) + Topic; Group × Topic models: Group (Polio, TB, no-prequestion) × Topic. Status, Condition, and Topic were coded with equal-prior contrasts, so the prior on the coefficients is also the prior on every pairwise difference between conditions (on the link scale). Other priors: intercepts normal(−1.5, 1) on the log scale (Gamma models; about 0.2 s per word), normal(0, 1.5) on the log-odds scale, and normal(0.2, 0.2) s per word for μ; ex-Gaussian σ and τ intercepts normal(−2.5, 1) on the softplus scale; random-effect *SD*s Student-*t*(3, 0, *ν*) with $\nu = 0.5$ (Gamma models, ex-Gaussian σ and τ), 1 (logistic and beta-binomial models), or 0.1 (ex-Gaussian μ); random-effect correlations LKJ(2); the Gamma shape gamma(2, 0.5); beta-binomial overdispersion exponential(0.1). Durations were modelled in s per word.